

Evolution of Rules in Reddit Communities

Harita Reddy

haritar2@illinois.edu

University of Illinois at Urbana-Champaign

Eshwar Chandrasekharan

eshwar@illinois.edu

University of Illinois at Urbana-Champaign

ABSTRACT

Reddit communities (*subreddits*) govern themselves through rules. Existing work does not study how rules change within a subreddit, and when different types of rules are added since subreddit creation. As current datasets lack information about subreddits' past rules, we present an approach to get all rules in the history of a subreddit using the Wayback Machine, and study rule trajectories within 496 subreddits. We find that at least 50% of the first rule additions occurred in less than a year from subreddit creation. Rules about civility and relevance are most common in the first rules. In contrast, rules that prohibit certain content are frequently added over time. Studying rule trajectories can help us answer whether early rule establishment helps with community success, improving key factors like user retention, norm compliance, and overall desirable behavior within the subreddit.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**; **Empirical studies in collaborative and social computing**.

KEYWORDS

online governance; content moderation; rule creation; rule changes

ACM Reference Format:

Harita Reddy and Eshwar Chandrasekharan. 2023. Evolution of Rules in Reddit Communities. In *Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*, October 14–18, 2023, Minneapolis, MN, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3584931.3606973>

1 INTRODUCTION

Reddit is organized into communities called *subreddits* which allow users to engage in discussion by creating posts or submissions, and commenting on them [1]. Subreddits are typically moderated by groups of volunteer moderators who enforce subreddit rules and norms. Prior work has shown that there exist norms at *macro*, *meso*, and *micro* levels across subreddits [3]. Rules, unlike implicit norms, are a formalized list of guidelines, typically present on a subreddit's homepage and are managed by moderators. Explicitly stated rules explaining acceptable behavior are easier to grasp when compared

to implicit norms, particularly for newcomers [5, 7]. Popular subreddits have a higher probability of having rules [5]. Rules can be prescriptive or restrictive and may be categorized based on their target behavior (e.g., hate speech and harassment) [5]. Through interviews with moderators, research has shown that unexpected incidents led to introduction of new rules—e.g., curb sudden influx of bad behavior, and adhere to new site-wide policies [9].

1.1 Summary of Methods and Contributions

Background: While prior work has focused on studying what types of norms and rules have been established in communities [3, 5], we lack information about how rules are created during the initial stages of a subreddit, and how rules evolve over time. The two key gaps that we try to address through our work include compiling a dataset of rule changes over time within subreddits, and studying how rule (addition) trajectories look like within newly-created subreddits. We specifically look at how early subreddits establish their first or initial rules, how often they add and modify rules at different stages of their life, and what type of rules they add at which stage of their life. This work has implications for understanding how early subreddits should start formalising rules in order to grow in a sustainable manner, and whether frequent updating and modification of rules is beneficial for community members.

Methods and Findings: We present an approach to get all rules in the history of a subreddit, and determine rule changes to understand the historical progression of rules over time. We group 600 subreddits into three cohorts of equal size based on their creation years (2019, 2020, 2021). Only 496 out of the 600 had at least one rule added during the period of evaluation. At least half of the first rule additions took place in less than a year since subreddit creation, thus showing the adoption of rules widget for formally stating rules. We then look at whether the number of rule changes drop as the age of the cohort increases. We note that while most rule additions occur during the initial stages of life of the cohorts, rule modifications do not follow a very consistent pattern across cohorts. Finally, we manually label rules to observe the type of rules being added in different rounds of rule additions within a subreddit. We note that most subreddits add rules pertaining to civility and topic-relevance in their first rule additions, showing the importance of such rules from inception. In later rule additions, rules prohibiting certain types of content are more common, indicating that subreddits keep adding new rules with time to prevent certain types of content for maintaining overall quality.

Implications: We present a large-scale study of rule changes across time within 496 subreddits since their creation. Our methodology can be extended in the future to examine how rule changes over time affect the health and success of communities. In our future work, we plan to study what types of rules are important to be added at specific stages of life to increase user retention and

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CSCW '23 Companion, October 14–18, 2023, Minneapolis, MN, USA

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ACM ISBN 979-8-4007-0129-0/23/10...\$15.00

<https://doi.org/10.1145/3584931.3606973>

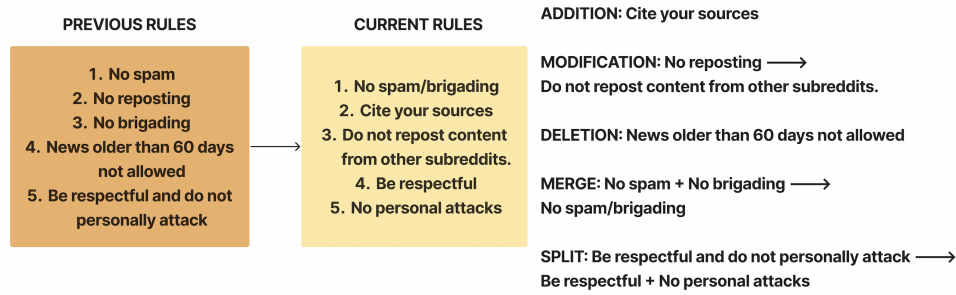


Figure 1: An example of 5 types of rule changes happening in between two consecutive snapshots.

diversity, norm compliance, and other measures of community success. This research is beneficial for moderators who can leverage our findings to ensure that they are setting up their subreddits for success by introducing different types of rules when required. Recommendations for the types of rules to be added could be incorporated into the Reddit interface (e.g., in the form of sensible defaults) to assist moderators of emerging subreddits.

2 METHODOLOGY

2.1 Recreating Historical Snapshots of Subreddits Using Wayback Machine

Commonly used sources of historical Reddit data lack the information about past rules. PRAW, a wrapper for Reddit API, can only be used to fetch subreddit's current metadata¹. Pushshift stores past posts and comments, and some metadata such as the number of subscribers at the time of posting [2]. However, as per our knowledge, it does not contain any information about rules of the subreddits in the past. Therefore, we turn to Wayback Machine to get the historical snapshots of the homepages of subreddits [6, 8]. Recent work used a similar approach to extract subreddit rules [6]. We get the snapshots of the homepages of the selected subreddits till June 2022, which is the end of our *evaluation period*. The limitation of using Wayback Machine is that the number of snapshots available for different subreddits can vary significantly. While it is noticeable that highly popular subreddits have snapshots saved almost daily, there exist subreddits with sparsely scattered snapshots, with gaps between two consecutive snapshots even going beyond an year.

2.2 Grouping Subreddits into Cohorts Based on Creation Year

We compile a list of subreddits created in 2019, 2020 and 2021 using two public datasets of subreddit names and creation dates²³. We group subreddits created in the same year into the same cohort and dropped subreddits that had fewer than 20 Wayback Machine snapshots. Finally we sampled 200 subreddits each for the cohort years 2019, 2020, and 2021, and collected their historical snapshots. Out of the 600 subreddits collected in total for the 3 cohorts, only 496 ever had any rules added. Our script uses regex to find the rules

widget and extract the rule headers or titles from the snapshots which use a rules widget. We chose to focus on the widget instead of rules appended to the description (in older version of Reddit) due to challenges in separating rules from other parts of the description.

2.3 Determining Rule Changes in Subreddits with Time

Rule changes at any point of time in history T are determined by comparing the list of rules in the snapshot at the nearest available prior timestamp t ($t < T$) with the list of rules in the snapshot at time T . We define 5 types of rule changes that can occur between two consecutive snapshots: (1) Rule Addition, (2) Rule Deletion, (3) Rule Modification, (4) Rule Merge, and (5) Rule Split. An example of all 5 types of rule changes is shown in figure 1.

Manual determination of rule changes by comparing consecutive snapshots is infeasible for analysis of a large number of subreddits. Instead, we opt for an automated technique to determine if rules have changed. We process snapshots of each subreddit in a chronological order, comparing rules at the current timestamp with the rules at the previous timestamp to determine any changes. We refer to rules in the previous snapshot as *old rules* and rules in the current snapshot as *new rules*. If an old rule and a new rule are exactly same, then no change has occurred for this rule. For the remaining rules, we calculate the similarity score between each old rule and each new rule. If no similarity-based match is found for the new rule, we assume that the new rule has been added. If they are slightly different, it means that the old rule has been modified to create the new rule. Since we are also considering splits and merges, we need to allow matching of more than one old rule to a single new rule, and more than one new rule to a single old rule in some situations (as seen in figure 1). We therefore allow multiple matching with some restrictions. We process each pair of old rule O_i and new rule N_j to find their similarity score. Rule pairs (O_i, N_j) where words in one of the rules are a subset of words in the other rule, are considered to be the pairs with highest similarity score, called as *subset matches*. The similarity between other pairs is computed using cosine similarity, with some additional weight given to whether O_i and N_j are present at the same index in the rule list. We process the pairs with a similarity score above a certain threshold which was determined based on the quality of results on a subset of rule changes. The processing is done in descending

¹<https://praw.readthedocs.io/en/stable/index.html>

²https://github.com/pakodanomics/list_of_128k_subreddit_names_nov2022

³<https://console.cloud.google.com/bigquery?p=fh-bigquery&page=project>

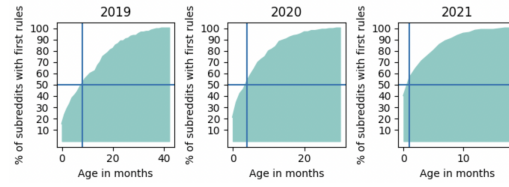


Figure 2: Lower bound of percentage of subreddits with their first rules added vs age in months. The lines mark the age by which 50% of the subreddits had their first rules added. When measuring percentage, we exclude subreddits which never had rules added during the evaluation period. 50% of the subreddits in each cohort had their first rules added before an year of age.

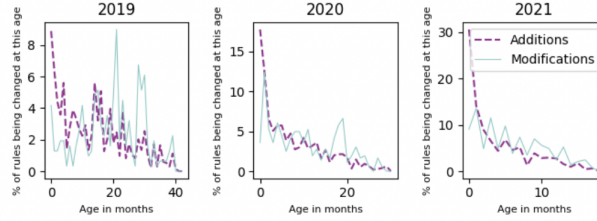


Figure 3: Percentage of rule additions and modifications at occurring at different ages in a cohort. Percentage of new rule additions occurring decreases with cohort’s age, but rule modifications do not show a consistent pattern.

order of the score to determine whether each pair (O_i, N_j) can be considered a match:

(1) If O_i has already been merged into a different new rule or N_j was created due to a split from a different old rule, we will eliminate the possibility of considering (O_i, N_j) as a match.

(2) If O_i has already been matched with other old rules, we check if at least one of those matches is a *subset match*. If not, we will not consider (O_i, N_j) as a match. In addition, if the difference between the similarity score of those existing matches and the similarity score of (O_i, N_j) is above a specific threshold, we will not consider (O_i, N_j) as a match. The threshold was determined based on manual observation of results on a few examples. These additional restrictions are placed to minimise bad cases of multiple matching (splits and merges) from occurring since similarity score calculation is not as accurate as humans determining similarity.

If a match is a one-to-one match between an old rule and a new rule, the change is considered to be a *modification*. If it is a one-to-many match from an old rule to many new rules, it is considered a *split*. If vice versa, it is considered a *merge*. Any unmatched old rules are considered *deletions* and unmatched new rules are considered to be *additions*.

3 FINDINGS AND DISCUSSION

3.1 When Are the First Rules Added to Subreddits?

Subreddits may not have rules right from the time of creation. Figure 2 shows the percentage of subreddits that had their first rules added by each age in months, and the point where 50% of the subreddits had their first rules added. When calculating this percentage, we exclude subreddits which never had rules added during the evaluation period. By *first rules*, we refer to the rules present in the

earliest snapshot of a subreddit with one or more rules. We note that 50% of the subreddits in each cohort had their first rules added within one year from creation. For the 2021 cohort, at least 50% of the first rule addition events occur before 2 months of age. We expect that this is due to the rules widget becoming more established and widely used and the newly created subreddits making use of it in their initial months itself. We also expect that subreddits which eventually became large in terms of number of subscribers would have had a smaller gap between their creation date and their first rule creation date. The reason behind our expectation is that subreddits that started controlling the quality of content through rules early would have been able to get more subscribers. To test this, we used the Pearson Correlation Coefficient [4]. As expected, we observe a negative correlation between the current subreddit subscriber count and the gap between their creation date and the first rule establishment, but the coefficient is small, with an average value of -0.15 across all cohorts.

3.2 Does Rule Change Activity Vary with Subreddit Age and Size?

Most rule additions occurring within a cohort are concentrated during the initial stages of subreddit’s life cycle. This is observed in Figure 3, where the percentage of rule additions occurring at a particular age keeps dropping as the age increases. This suggests that most rules are added during the initial months. Rule modifications do not show such a consistent pattern across all cohorts, with them peaking after the age of 20 months in the 2019 cohort. In 2020 and 2021 cohorts, we observe a pattern wherein both rule additions and rule modifications decrease with age. We also observe a positive correlation between the current subreddit subscriber count and the number of rule changes that happened over time during the

Table 1: Codes for manual labeling. Our interpretation of Fiesler et al’s codes could be different from what they intended [5]

Codes Proposed by Fiesler et al. [5]	Advertising & Commercialization, Harassment, Hate Speech, Images, Politics, Format, Content/Behavior, Copyright/Piracy, Doxxing/Personal Info, Links & Outside Content, Consequences/Moderation/Enforcement, Low-Quality Content, Off-topic, Personal Army, Personality, Reddiquette/Sitewide, Reposting, Spam, Spoilers, Trolling, Voting, NSFW
Additional Codes	Misleading Content/Misinformation, General Civility, Unsafe Content/Conduct, Prohibited Content

Table 2: For few labels, we show the % of additions that occurred in the first shot, i.e, in the first set of rules added to subreddits

Label	% of additions in the first shot
General Civility	82.84%
NSFW	78.02%
Hate Speech	77.42%
Prohibited Content	48.6%
Personality	47.78%

evaluation period, with an average value of 0.39 across the three cohorts.

3.3 What Types of Rules Are Added First and Across Time?

We used manual coding to label rules (as shown in Table 1) that were *first added* to subreddits in the 2019, 2020 and 2021 cohorts. By *first added rules*, we refer to the rules that are present in the first available snapshot of a subreddit in which there are non-zero number of rules. Due to lack of availability of snapshots between the creation date and the first available snapshot for a subreddit, we *assume* that the rules present in the first available snapshot are the *first added rules*. We used the existing codes proposed by Fiesler et al. and added our own custom codes based on what we felt was needed when manually labeling the rules [5]. Note that a single rule can be assigned multiple codes.

Most first added rules deal with Content and Behavior, leading to a large percentage for *Content/Behavior* label. More than 50% of the subreddits in all three cohorts have rules pertaining to *General Civility* in their first set of rules, thus indicating that ensuring people are nice and civil to each other is important for a significant number of subreddits at the beginning itself. In 2019 and 2020 cohorts, we also observe *Off-topic* related rules to be present in the first rules in at least 50% of the subreddits, thus showing the importance of specifying what topics are within the scope of a subreddit.

We then labeled all the rules that were added later in the course of life for the subreddits in the 2019, 2020, 2021 cohorts. Overall, we noticed some interesting patterns. As expected, there was a sudden increase in the number of additions of *Misleading Content/Misinformation* related rules during the early half of 2020. We expected this because during that time, several Coronavirus (COVID-19) subreddits came up, which emphasized on preventing misinformation. In table 2, we show the percentage of rule additions that occurred in first rule addition of subreddits for each label type.

Table 3: For each round of addition, we show the most frequent label added in that round, excluding the *Content/Behavior* label

Round	Most popular label by round of addition
1	General Civility
2	Prohibited Content
3	Prohibited Content
4	Prohibited Content
5	Prohibited Content

We notice that majority of *General Civility* rules were added in the first rule addition itself, indicating the importance of this rule right from the time of the creation of a subreddit. *General Civility* relates to the top values of remembering the human and standards of behavior on *Rediquette*. *Personality*, and *Prohibited Content* have the lowest percentages, indicating that rules related to these labels are added over time. *Prohibited Content* not being concentrated in the first rule additions indicates that subreddits keep adding disallowed content to their rules as time progresses.

Table 3 shows the most popular rule label by count in each round of addition. By *round of addition*, we are referring to the iteration of rule addition for a subreddit. If a label is added in the first rule addition in a subreddit, it is considered as a first round of addition. For example, let us suppose a subreddit adds rules R1 and R2 on April 1st, and then R3 is added on May 5, followed by R4 being added on Dec 2. R1 and R2 are said to be added in the first iteration or the first round of addition. R3 is said to be added in the second round of addition, and R4 is said to be added in the third round of addition. *General Civility* is the most commonly added rule type in the first round of addition. However, the addition of *Prohibited Content* related rules is more spread out, with it being the most frequent label added in rounds 2, 3, 4 and 5. Examples of rules labeled with *Prohibited Content* include rules such as *no looking for my buddy*, *we have a quest! posts*, and *no medical advice*. The pattern of addition of *Prohibited Content* label shows that moderators of the subreddits keep adding rules that prohibit certain types of content over time. One explanation could be that moderators created new rules to combat increases in specific types of content which was seen as undesirable within the community.

Limitations: A key limitation in the study is that the long gaps between snapshots make it difficult to identify the exact date on which the rule change event occurred. Also, the non-uniform frequency of snapshots for different subreddits make it difficult to compare two subreddits that have very different snapshot frequencies. To

overcome this, we should track subreddits of interest from their inception by regularly saving their home pages using the Wayback Machine.

4 CONCLUSION AND FUTURE DIRECTIONS

Our findings show that subreddits are willing to adopt the rules widget for explicitly stating rules and subreddits have a tendency to add rules related to civility and topic relevancy during their initial stages of life. In contrast, rules that prohibit certain types of content from being posted are added over time. Given the importance of civility and relevance from the creation time, newly created subreddits could be initialised with those rules as a default. One future step will be to study the relative success of subreddits with such rules in the beginning. Also, given that subreddits keep adding rules prohibiting certain types of content, they could keep a running checklist of disallowed content in one place. The checklist could be possibly shown on the interface which appears when a user is creating a new post, thus improving users' awareness of what is prohibited. Furthermore, our study could be extended to examine the causal effects of a new rule's addition on eventual community-level outcomes like user retention, norm compliance, diversity of user base, uniqueness of content generated, and so on. Based on such insights, Reddit could automatically recommend certain types of rules to be added within a subreddit in order to promote norm compliance (especially in response to sudden spikes in certain behaviors) or improve the retention of newcomers.

REFERENCES

- [1] Katie Elson Anderson. 2015. Ask me anything: what is Reddit? *Library Hi Tech News* 32, 5 (2015), 8–11.
- [2] Jason Baumgartner, Savvas Zannettou, Brian Keegan, Megan Squire, and Jeremy Blackburn. 2020. The pushshift reddit dataset. In *Proceedings of the international AAAI conference on web and social media*, Vol. 14. 830–839.
- [3] Eshwar Chandrasekharan, Mattia Samory, Shagun Jhaver, Hunter Charvat, Amy Bruckman, Cliff Lampe, Jacob Eisenstein, and Eric Gilbert. 2018. The Internet's hidden rules: An empirical study of Reddit norm violations at micro, meso, and macro scales. *Proceedings of the ACM on Human-Computer Interaction* 2, CSCW (2018), 1–25.
- [4] Israel Cohen, Yiteng Huang, Jingdong Chen, Jacob Benesty, Jacob Benesty, Jingdong Chen, Yiteng Huang, and Israel Cohen. 2009. Pearson correlation coefficient. *Noise reduction in speech processing* (2009), 1–4.
- [5] Casey Fiesler, Jialun Jiang, Joshua McCann, Kyle Frye, and Jed Brubaker. 2018. Reddit rules! characterizing an ecosystem of governance. In *Proceedings of the International AAAI Conference on Web and Social Media*, Vol. 12.
- [6] Seth Frey, Qiankun Zhong, Beril Bulat, William D Weisman, Caitlyn Liu, Stephen Fujimoto, Hannah Wang, and Charles M Schweik. 2022. Governing online goods: Maturity and formalization in Minecraft, Reddit, and World of Warcraft communities. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–23.
- [7] Cliff Lampe, Paul Zube, Jusil Lee, Chul Hyun Park, and Erik Johnston. 2014. Crowdsourcing civility: A natural experiment examining the effects of distributed moderation in online forums. *Government Information Quarterly* 31, 2 (2014), 317–326.
- [8] Jamie Murphy, Noor Hazarina Hashim, and Peter O'Connor. 2007. Take me back: validating the wayback machine. *Journal of Computer-Mediated Communication* 13, 1 (2007), 60–75.
- [9] Joseph Seering, Tony Wang, Jina Yoon, and Geoff Kaufman. 2019. Moderator engagement and community development in the age of algorithms. *New Media & Society* 21, 7 (2019), 1417–1443.